

**Aufgabe 1:** (24 Punkte)

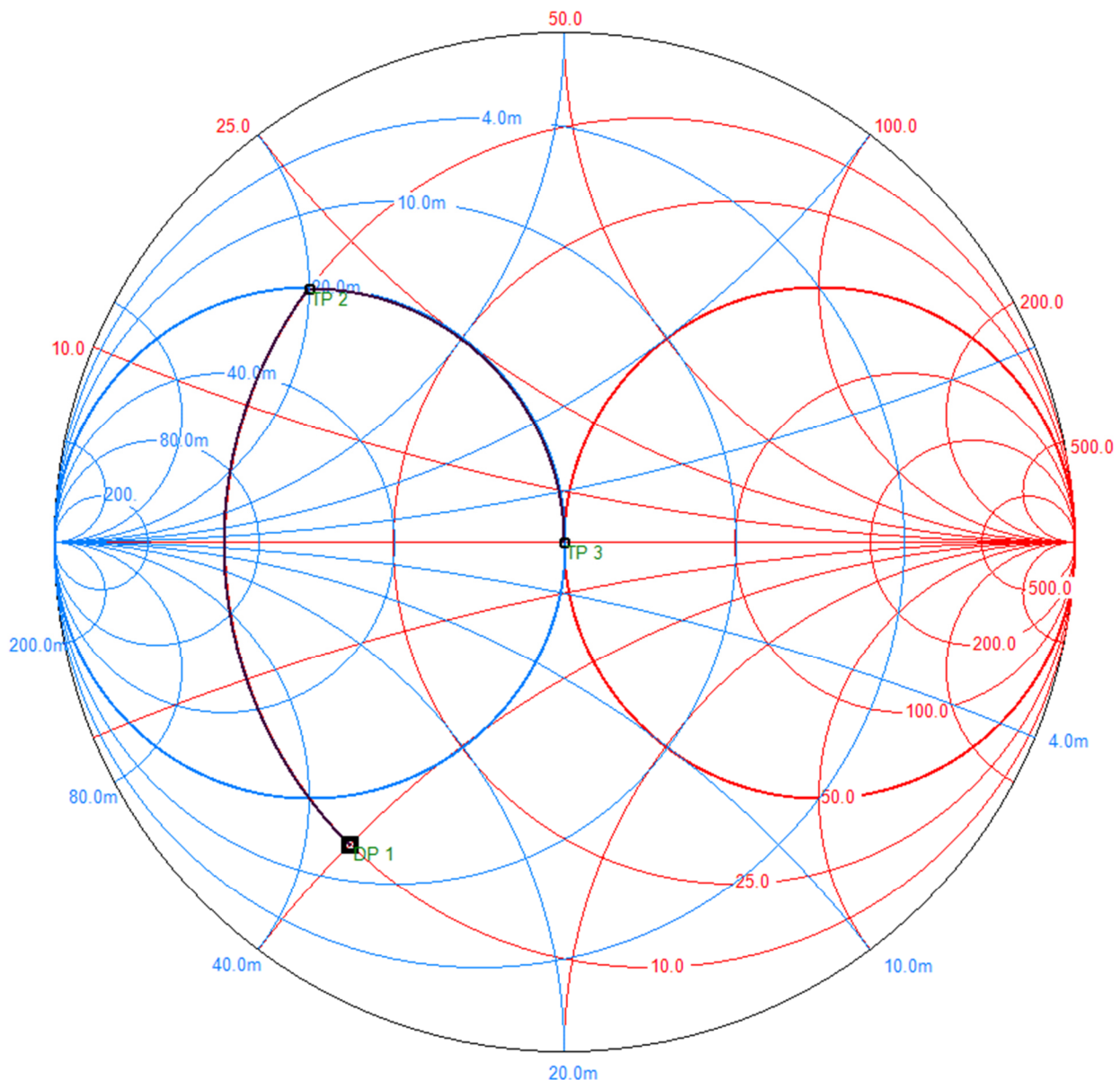
1.

$$\underline{Z}_{e,Tr} = Z_L \cdot \frac{1 + \underline{s}_{11}}{1 - \underline{s}_{11}} = 50\Omega \cdot \frac{1 + 0,7257 \cdot e^{-j125,4^\circ}}{1 - 0,7257 \cdot e^{-j125,4^\circ}} = (10 - j25)\Omega$$

$$\underline{Z}_{e,Tr} \text{ normiert: } \frac{\underline{Z}_{e,Tr}}{Z_L} = \frac{(10 - j25)\Omega}{50\Omega} = 0,2 - j0,5$$

$\underline{Z}_e$  ins Smith-Chart eintragen (oder  $\underline{s}_{11}$  mit Betrag und Phase direkt ins Smith Chart).

Um Längs-L transformieren  $\rightarrow$  Punkt 2; Um Parallel-C transformieren  $\rightarrow 50\Omega$



ablesen:  $x_L = 0,40 - (-0,50) = 0,90$      $b_C = 0,0 - (-2,0) = 2,0$

$$\underline{L} = \frac{x_L \cdot Z_L}{\omega} = \frac{0,90 \cdot 50 \frac{\text{V}}{\text{A}}}{2\pi \cdot 2,5 \cdot 10^9 \frac{1}{\text{s}}} = \underline{\underline{2,86 \text{ nH}}}$$

$$\underline{C} = \frac{b_C}{Z_L \cdot \omega} = \frac{2,0}{50 \frac{\text{V}}{\text{A}} \cdot 2\pi \cdot 2,5 \cdot 10^9 \frac{1}{\text{s}}} = \underline{\underline{2,55 \text{ pF}}}$$

2.

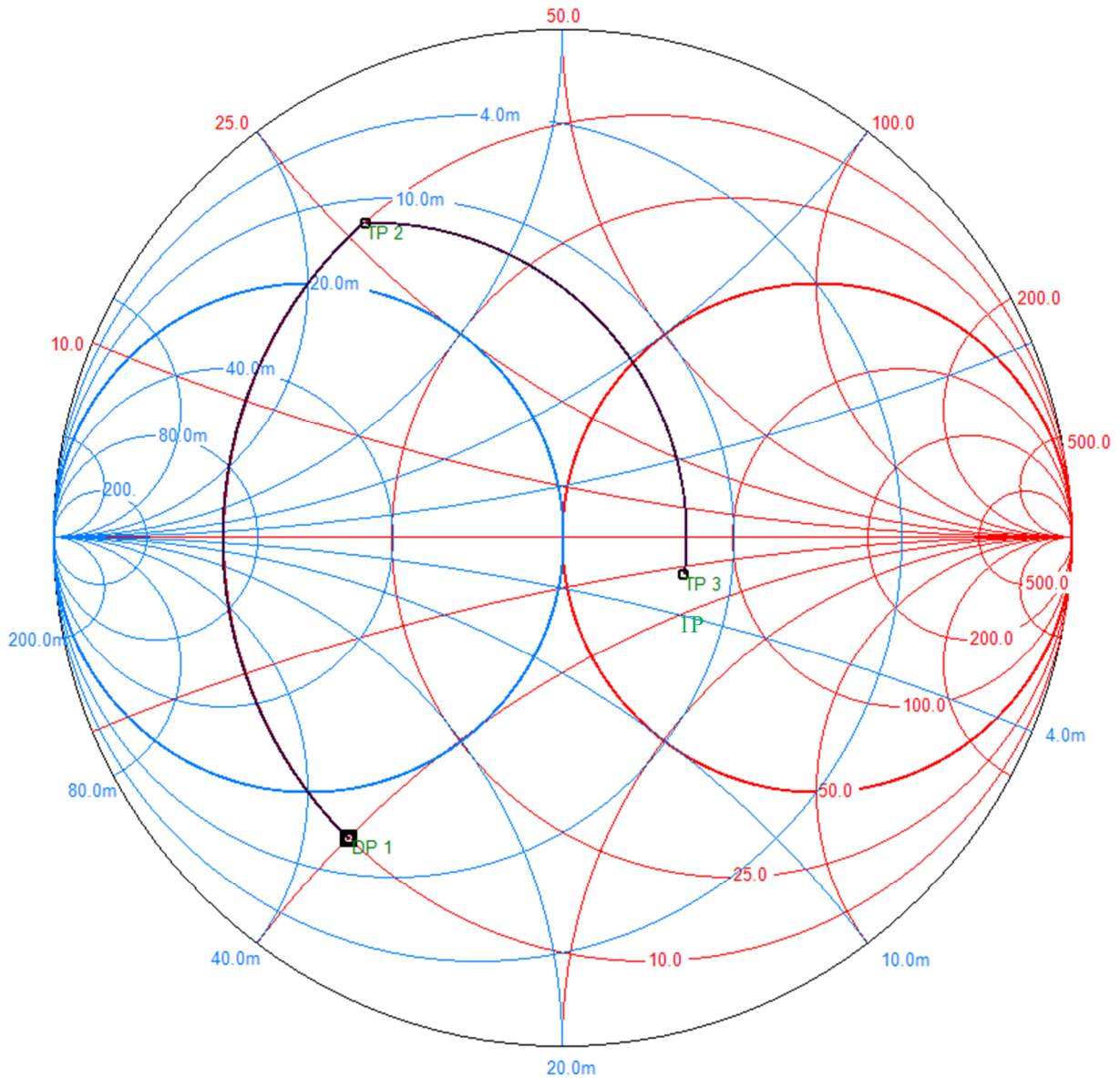
$$X_L = \omega \cdot L = 2\pi \cdot 2,5 \cdot 10^9 \frac{1}{s} \cdot 3,3 \cdot 10^{-9} \frac{Vs}{A} = 51,84\Omega$$

$$x_L = \frac{X_L}{Z_L} = \frac{51,84\Omega}{50\Omega} = 1,037$$

$$X_C = -\frac{1}{\omega \cdot C} = \frac{-1}{2\pi \cdot 2,5 \cdot 10^9 \frac{1}{s} \cdot 2,2 \cdot 10^{-12} \frac{As}{V}} = -28,94\Omega$$

$$b_C = -\frac{Z_L}{X_C} = -\frac{50\Omega}{-28,94\Omega} = 1,728$$

über Smith-Chart:



$$x_{TP2} = 1,037 + (-0,5) = 0,537$$

$$b_{TP2} = -1,634$$

$$b_{TP3} = -1,634 + 1,728 = 0,094$$

$$\text{im Smith-Chart abgemessen: } |r_{TP3}| = 0,25 \quad \underline{\underline{a_r = 12,0\text{dB}}}$$

$$\underline{Z}_e = \frac{(\underline{Z}_{e,Tr} + jX_L) \cdot jX_C}{(\underline{Z}_{e,Tr} + jX_L) + jX_C} = \frac{((10 - j25)\Omega + j51,84\Omega) \cdot (-j28,94\Omega)}{(10 - j25)\Omega + j51,84\Omega - j28,94\Omega} = (80,21 - j12,09)\Omega$$

$$r_{\underline{e}} = \frac{\underline{Z}_e - Z_L}{\underline{Z}_e + Z_L} = \frac{(80, 21 - j12, 09\Omega) - 50\Omega}{(80, 21 - j12, 09\Omega) + 50\Omega} = 0,249 \cdot e^{-j16,5^\circ}$$

$$\underline{\underline{a_r = 20\text{dB} \cdot \lg\left(\frac{1}{0,2489}\right) = 12,1\text{dB}}}$$

- $$\lambda = \frac{c_0}{\sqrt{\varepsilon_r} \cdot f} = \frac{3 \cdot 10^8 \frac{\text{m}}{\text{s}}}{\sqrt{3,3} \cdot 2,5 \cdot 10^9 \frac{1}{\text{s}}} = 66,06 \text{mm} \quad \frac{l_{\min}}{\lambda} = \frac{5,0 \text{mm}}{66,06 \text{mm}} = 0,0757$$

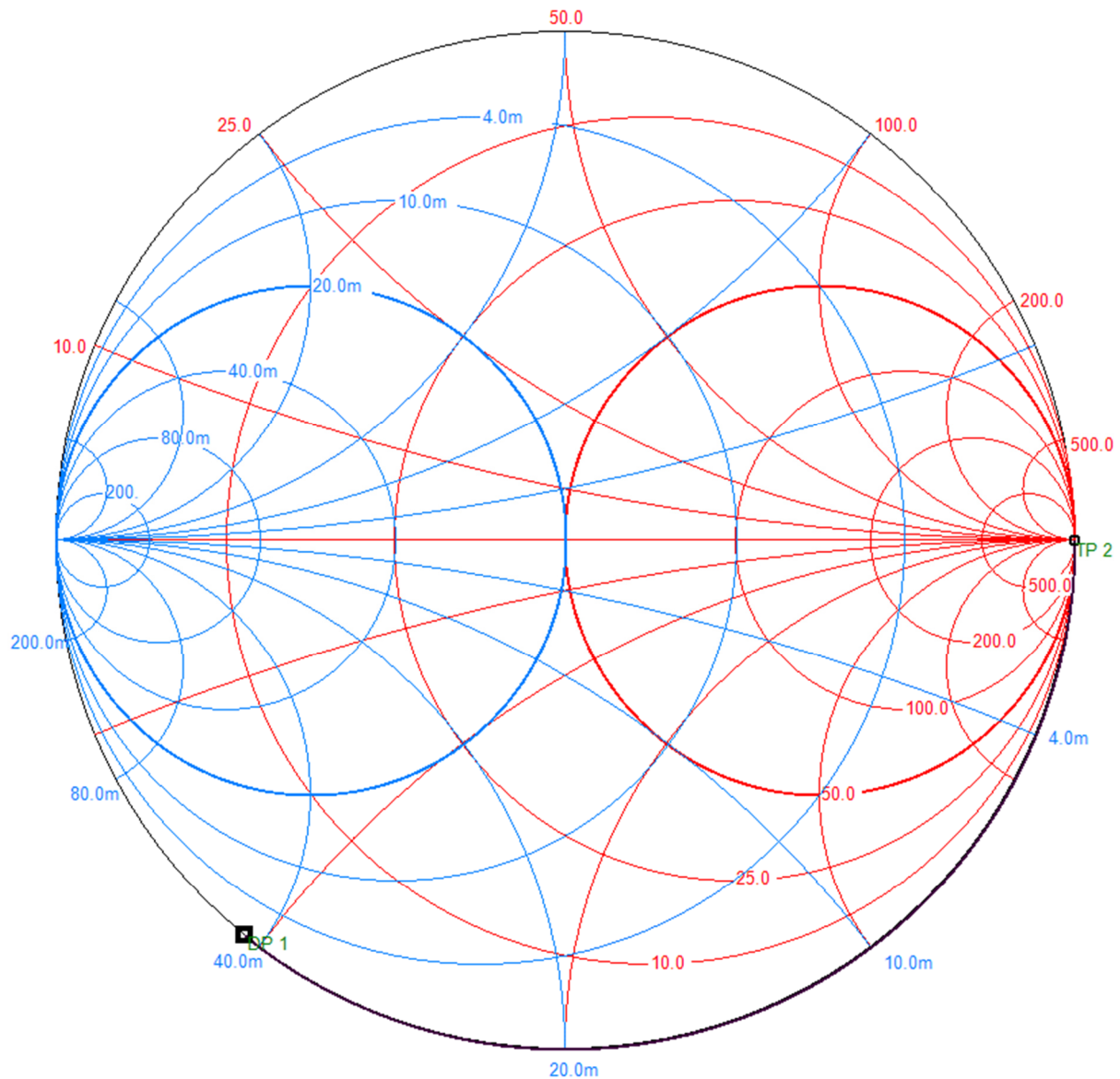
$$\frac{l}{\lambda} = 0,438 - 0,424 = 0,014 < 0,0757 = \frac{l_{min}}{\lambda}$$
[illegible]

aus Smith-Chart abgelesen:

$$\frac{l_l}{\lambda} = 0,061 - 0,424 + 0,5 = 0,137 \quad \rightarrow \quad \underline{\underline{l_l = 0,137 \cdot 66,06\text{mm} = 9,05\text{mm}}}$$

$$b_{\text{Stich}} = 0 - (-2,1) = 2,1$$

Stichleitung muss kapazitiv sein und kleiner  $\lambda/4 \rightarrow$  leerlaufende Stichleitung



aus Smith-Chart abgelesen:

$$\frac{l_s}{\lambda} = 0,428 - 0,25 = 0,178 \quad \rightarrow \quad \underline{\underline{l_s = 0,178 \cdot 66,06\text{mm} = 11,8\text{mm}}}$$

**Aufgabe 2:**

(18 Punkte)

1. mit 3-6 für  $Z_L = 50\Omega$ :  $\frac{w}{h} = 2,1 \rightarrow \underline{\underline{w}} = 2,1 \cdot h = 2,1 \cdot 1,2\text{mm} = \underline{\underline{2,52\text{mm}}}$

mit 3-5 für  $\frac{w}{h} = 2,1$ :  $\varepsilon_{r,eff} = 2,05$

$$\underline{\underline{\lambda}} = \frac{c_0}{\sqrt{\varepsilon_{r,eff}} \cdot f} = \frac{3 \cdot 10^8 \frac{\text{m}}{\text{s}}}{\sqrt{2,05} \cdot 5,0 \cdot 10^9 \frac{1}{\text{s}}} = \underline{\underline{42\text{mm}}}$$

2.  $\frac{s}{h} = \frac{3,2\text{mm}}{1,2\text{mm}} = 2,7 \rightarrow k_L = 0,05$

$$\begin{aligned} |\underline{s}_{41}|^2 &= \frac{k_L^2 \cdot \sin^2 \beta l}{(1 - k_L^2) \cdot \cos^2 \beta l + \sin^2 \beta l} = \frac{k_L^2 \cdot \sin^2 \beta l}{-k_L^2 \cdot \cos^2 \beta l + \cos^2 \beta l + \sin^2 \beta l} = \frac{k_L^2 \cdot \sin^2 \beta l}{1 - k_L^2 \cdot \cos^2 \beta l} = \\ &= \frac{k_L^2 \cdot \sin^2 \left( 2\pi \cdot \frac{l}{\lambda} \right)}{1 - k_L^2 \cdot \cos^2 \left( 2\pi \cdot \frac{l}{\lambda} \right)} \approx k_L^2 \cdot \sin^2 \left( 2\pi \cdot \frac{l}{\lambda} \right) = 0,05^2 \sin^2 \left( 2\pi \cdot \frac{64\text{mm}}{42\text{mm}} \right) = 5,55 \cdot 10^{-5} \end{aligned}$$

$$\frac{P_E}{P_S} \triangleq 10\text{dB} \cdot \lg(5,55 \cdot 10^{-5}) = \underline{\underline{-42,6\text{dB}}}$$

3. keine Kopplung für:

$$l_0 = n \cdot \frac{\lambda}{2} = n \cdot \frac{42\text{mm}}{2} = n \cdot 21\text{mm}$$

für  $n = 3$ :  $\underline{\underline{l_0}} = 3 \cdot 21\text{mm} = \underline{\underline{63\text{mm}}} = l - 1\text{mm}$  ist im erlaubten Bereich!

**Aufgabe 3:**

(19 Punkte)

$$1. \quad \underline{\underline{Z_E}} = Z_L \cdot \frac{1 + \underline{s_{11}}}{1 - \underline{s_{11}}} = 50\Omega \cdot \frac{1 + 0,73 \cdot e^{-j75^\circ}}{1 - 0,73 \cdot e^{-j75^\circ}} = \underline{\underline{(20,2 - j61,0)\Omega}} = 64,3\Omega \cdot e^{-j71,7^\circ}$$

$$\underline{\underline{Z_A}} = Z_L \cdot \frac{1 + \underline{s_{22}}}{1 - \underline{s_{22}}} = 50\Omega \cdot \frac{1 + 0,82 \cdot e^{j13^\circ}}{1 - 0,82 \cdot e^{j13^\circ}} = \underline{\underline{(220 + j248)\Omega}} = 331\Omega \cdot e^{j48,4^\circ}$$

$$2. \quad \underline{\underline{V_{Tr}}} = 10\text{dB} \cdot \lg |\underline{s_{21}}|^2 = 10\text{dB} \cdot \lg(5,8^2) = \underline{\underline{15,27\text{dB}}}$$

$$\underline{\underline{V_{Tr,max}}} = 10\text{dB} \cdot \lg \left( |\underline{s_{21}}|^2 \cdot \frac{1}{1 - |\underline{s_{11}}|^2} \cdot \frac{1}{1 - |\underline{s_{22}}|^2} \right) = 10\text{dB} \cdot \lg \left( 5,8^2 \cdot \frac{1}{1 - 0,73^2} \cdot \frac{1}{1 - 0,82^2} \right) = 23,42\text{dB}$$

"mehr" an Verstärkung:  $\underline{\underline{\Delta V}} = V_{Tr,max} - V_{Tr} = 23,42\text{dB} - 15,27\text{dB} = \underline{\underline{8,15\text{dB}}}$

oder:

$$\left[ \Delta V = 10\text{dB} \cdot \lg \left( \frac{1}{1 - |\underline{s_{11}}|^2} \cdot \frac{1}{1 - |\underline{s_{22}}|^2} \right) = 8,15\text{dB} \right]$$

$$3. \quad B = 1 + |\underline{s_{11}}|^2 - |\underline{s_{22}}|^2 - |\underline{s_{11}}\underline{s_{22}} - \underline{s_{12}}\underline{s_{21}}|^2 \geq 0$$

$$\rightarrow |\underline{s_{11}}\underline{s_{22}} - \underline{s_{12,max}}\underline{s_{21}}| = \sqrt{1 + |\underline{s_{11}}|^2 - |\underline{s_{22}}|^2}$$

„ungünstigste Phasenlage“ heißt:  $\underline{s_{11}}\underline{s_{22}}$  und  $\underline{s_{12}}\underline{s_{21}}$  addieren sich zu  $|\underline{s_{11}}\underline{s_{22}}| + |\underline{s_{12}}\underline{s_{21}}|$ :

$$|\underline{s_{12,max}}| \cdot |\underline{s_{21}}| = \sqrt{1 + |\underline{s_{11}}|^2 - |\underline{s_{22}}|^2} - |\underline{s_{11}}\underline{s_{22}}|$$

$$\underline{\underline{|\underline{s_{12,max}}|}} = \frac{\sqrt{1 + |\underline{s_{11}}|^2 - |\underline{s_{22}}|^2} - |\underline{s_{11}}| \cdot |\underline{s_{22}}|}{|\underline{s_{21}}|} = \frac{\sqrt{1 + 0,73^2 - 0,82^2} - 0,73 \cdot 0,82}{5,8} = \underline{\underline{5,67 \cdot 10^{-2}}}$$

$$K = \frac{1 - |\underline{s_{11}}|^2 - |\underline{s_{22}}|^2 + \left| \underline{s_{11}}\underline{s_{22}} + \underline{s_{12,max}}\underline{s_{21}} \right|^2}{2 |\underline{s_{12,max}}\underline{s_{21}}|} =$$

$$= \frac{1 - 0,73^2 - 0,82^2 + (0,73 \cdot 0,82 + 5,8 \cdot 5,67 \cdot 10^{-2})^2}{2 \cdot 5,67 \cdot 10^{-2} \cdot 5,8} = 0,996 < 1$$

Transistor ist **nicht** absolut stabil!

**Aufgabe 4:**

(19 Punkte)

1.

$$F_{LNB} = 10^{\frac{0,5\text{dB}}{10\text{dB}}} = 1,12 \quad G_{LNB} = 10^{\frac{42\text{dB}}{10\text{dB}}} = 15,8 \cdot 10^3$$

$$G_L = 10^{\frac{-40\text{dB}}{10\text{dB}}} = 1,0 \cdot 10^{-4} \quad F_L = \frac{1}{G_L} = 10000$$

$$F_V = 10^{\frac{6,0\text{dB}}{10\text{dB}}} = 3,98 \quad G_V = 10^{\frac{20\text{dB}}{10\text{dB}}} = 100$$

$$F_{ges} = 1 + (F_{LNB} - 1) + \frac{F_L - 1}{G_{LNB}} + \frac{F_V - 1}{G_{LNB} \cdot G_L} = 1 + (1,12 - 1) + \frac{10000 - 1}{15,8 \cdot 10^3} + \frac{3,98 - 1}{15,8 \cdot 10^3 \cdot 1,0 \cdot 10^{-4}} = 3,64$$

$$\underline{\underline{f_{ges} = 10\text{dB} \cdot \lg F_{ges} = 5,61\text{dB}}}$$

2.

$$P_{RA} = k \cdot T_A \cdot B = 1,38 \cdot 10^{-23} \frac{\text{Ws}}{\text{K}} \cdot 38\text{K} \cdot 30 \cdot 10^6 \frac{1}{\text{s}} = 15,73 \cdot 10^{-15} \text{W} \hat{=} -108,0\text{dBm}$$

$$\left(\frac{S}{N}\right)_A = \frac{P_A}{P_{RA}} \hat{=} -90\text{dBm} - (-108,0\text{dBm}) = 18\text{dB}$$

$$\underline{\underline{\left(\frac{S}{N}\right)_{Rec} = \left(\frac{S}{N}\right)_A - f_{ges} = 18\text{dB} - 5,0\text{dB} = 13\text{dB}}}$$

3.

$$F'_{ges} = 1 + (F_{LNB} - 1) + \frac{F_L - 1}{G_{LNB}} = 1 + (1,12 - 1) + \frac{10000 - 1}{15,8 \cdot 10^3} = 1,75$$

$$f'_{ges} = 10\text{dB} \cdot \lg F'_{ges} = 2,4\text{dB}$$

$$\underline{\underline{\left(\frac{S}{N}\right)_{Rec, ohne\_V} = \left(\frac{S}{N}\right)_A - f'_{ges} = 18\text{dB} - 2,4\text{dB} = 15,6\text{dB}}} \quad [17,6\text{dB}]$$

**Aufgabe 5:**

(20 Punkte)

$$1. \quad P_{Q,max} = \frac{1}{2} |\underline{b}_Q|^2 \cdot \frac{1}{1 - |\underline{r}_Q|^2}$$

$$\underline{\underline{P_Q}} = \frac{1}{2} |\underline{b}_Q|^2 = P_{Q,max} \cdot (1 - |\underline{r}_Q|^2) = 10 \text{ W} \cdot (1 - 0,8^2) = \underline{\underline{3,6 \text{ W}}}$$

$$2. \quad \underline{r}_L = \underline{r}_2^* \quad \text{für Anpassung}$$

$$\underline{r}_2 = \underline{s}_{22} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_Q}{1 - \underline{s}_{11} \cdot \underline{r}_Q} = j0,6 + \frac{0,8 \cdot 0,8 \cdot (-j0,8)}{1 - j0,6 \cdot (-j0,8)} = j0,6 + \frac{-j0,512}{1 - 0,48} = -j0,3846$$

$$\rightarrow \underline{\underline{r_{L,opt}}} = j0,3846$$

$$\underline{b}_{Q2} = \underline{b}_Q \cdot \frac{\underline{s}_{21}}{1 - \underline{s}_{11} \cdot \underline{r}_Q} = \underline{b}_Q \cdot \frac{0,8}{1 - j0,6 \cdot (-j0,8)} = \underline{b}_Q \cdot 1,538$$

$$\underline{\underline{P_{L,max}}} = \frac{1}{2} |\underline{b}_{Q2}|^2 \cdot \frac{1 - |\underline{r_{L,opt}}|^2}{|1 - \underline{r}_2 \cdot \underline{r_{L,opt}}|^2} = 3,6 \text{ W} \cdot 1,538^2 \cdot \frac{1 - 0,3846^2}{|1 - j0,3846 \cdot (-j0,3846)|^2} = \underline{\underline{10 \text{ W}}} \quad [8,33 \text{ W}]$$

$$3. \quad \underline{r}_1 = \underline{s}_{11} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r_{L,opt}}}{1 - \underline{s}_{22} \cdot \underline{r_{L,opt}}} = j0,6 + \frac{0,8 \cdot 0,8 \cdot j0,3846}{1 - j0,6 \cdot j0,3846} = j0,6 + \frac{j0,2461}{1 + 0,2308} = j0,8000$$

$$P_{Q,ab} = \frac{1}{2} |\underline{b}_Q|^2 \cdot \frac{1 - |\underline{r}_1|^2}{|1 - \underline{r}_Q \cdot \underline{r}_1|^2} = 3,6 \text{ W} \cdot \frac{1 - 0,8000^2}{|1 - (-j0,8) \cdot j0,8000|^2} = 10 \text{ W} \quad [8,33 \text{ W}]$$

$$\underline{\underline{a_B}} = 10 \text{ dB} \cdot \lg \left( \frac{P_{Q,ab}}{P_{L,max}} \right) = 10 \text{ dB} \cdot \lg \left( \frac{10 \text{ W}}{10 \text{ W}} \right) = \underline{\underline{0,00 \text{ dB}}}$$

ohne 2-Tor:

$$P_{L,abs0} = \frac{1}{2} |\underline{b}_Q|^2 \cdot \frac{1 - |\underline{r_{L,opt}}|^2}{|1 - \underline{r}_Q \cdot \underline{r_{L,opt}}|^2} = 3,6 \text{ W} \cdot \frac{1 - 0,3846^2}{|1 - (-j0,8) \cdot j0,3846|^2} = 6,40 \text{ W} \quad [5,33 \text{ W}]$$

$$\underline{\underline{a_E}} = 10 \text{ dB} \cdot \lg \left( \frac{P_{L,abs0}}{P_{L,max}} \right) = 10 \text{ dB} \cdot \lg \left( \frac{6,40 \text{ W}}{10 \text{ W}} \right) = \underline{\underline{-1,9 \text{ dB}}}$$